📊 Raport z klasyfikacji bankotów obiegowych

1️⃣Dane

359 zdjęć
6 klas - (51, 57, 62, 65, 65, 59) 10 zloty,20 zloty,50 zloty,100 zloty,200 zloty,500 zloty

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2️⃣Testowanie hiperparametrów

Szybkość treningu

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "TrainingSpeed",
    "task_name":        "CPU_NoTransfer",
    "img_size":         (32, 32),
    "batch_size":       16,
    "epochs":           10,
    "learning_rate":    0.001,
    "model_type":       "resnet",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      False

}
```

TYPE	NAME	TAGS	STATUS	USER	STARTED	UPDATED	ITERATI...	PARENT TASK	RUN TIME
Training	task_name: GPU_NoTransfer img_size: (32, 32) batch_size:...		Completed	Marcel Raś	2 days ago	2 days ago	9		02:52m
Training	task_name: CPU_NoTransfer img_size: (32, 32) batch_size: ...		Completed	Marcel Raś	3 days ago	3 days ago	9		08:16m

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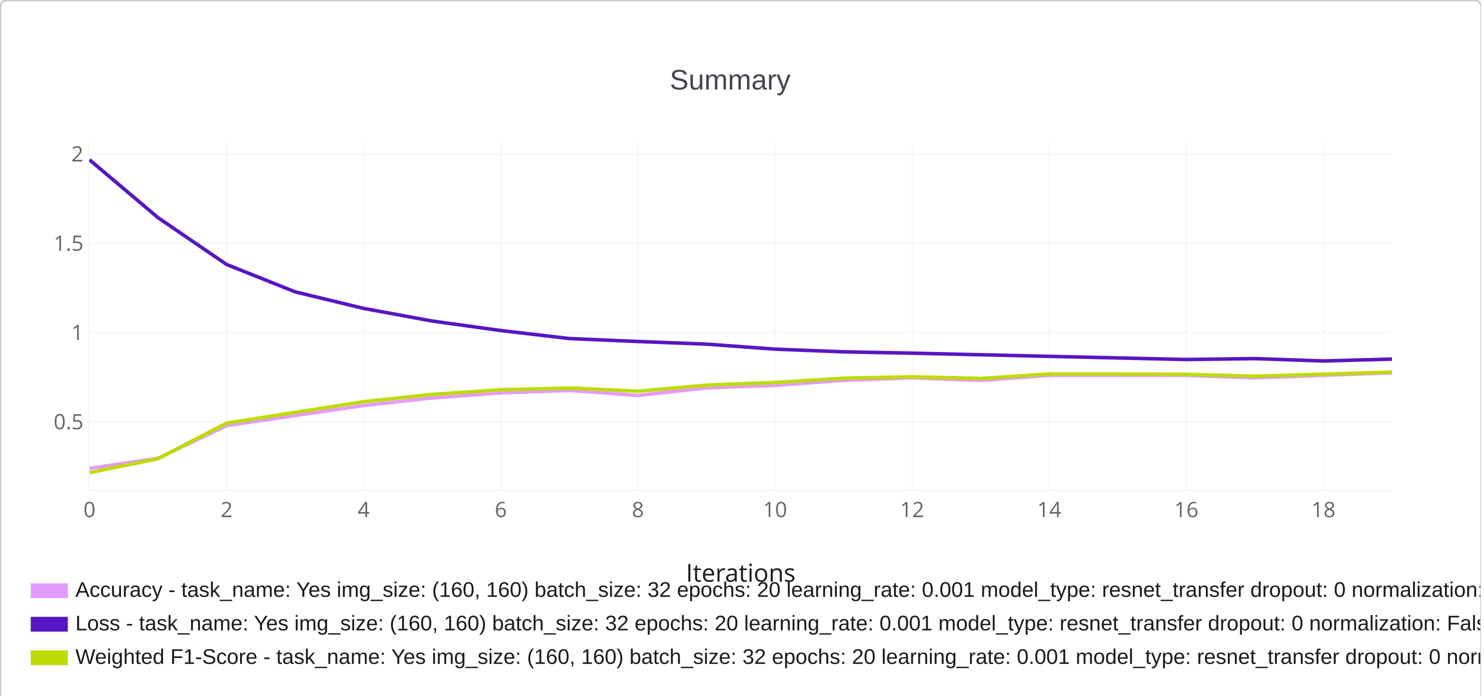
TransferLearning

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "TransferLearning",
    "task_name":        "No",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      False

}
```

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "TransferLearning",
    "task_name":        "Yes",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      False

}
```



Testowanie Sieci

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Models",
    "task_name":        "",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      False

}
models = ['resnet_transfer', 'resnet', 'vgg16', 'mobilenet', 'inception']
for model in models:
  cfg['model_type'] = model
  cfg['task_name'] = model
  cfg["task_name"] = generate_taskname(cfg)
  start(cfg)
```

Augmentacja

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Augment",
    "task_name":        "",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      False

}


augment = [0.0,0.1,0.2,0.4]
for augments in augment:
  cfg['augment'] = augments
  cfg['task_name'] = f"Augment: {augments}"
  cfg["task_name"] = generate_taskname(cfg)
  start(cfg)
```

Normalizacja

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Normalize",
    "task_name":        "No",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
```

```
    'add_newdata':      False

}
cfg["task_name"] = generate_taskname(cfg)
start(cfg)

cfg = {
    "project_root":     "Projekt",
    "subproject":       "Normalize",
    "task_name":        "Yes",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    True,
    'augment':          0,
    'add_newdata':      False

}
cfg["task_name"] = generate_taskname(cfg)
start(cfg)
```

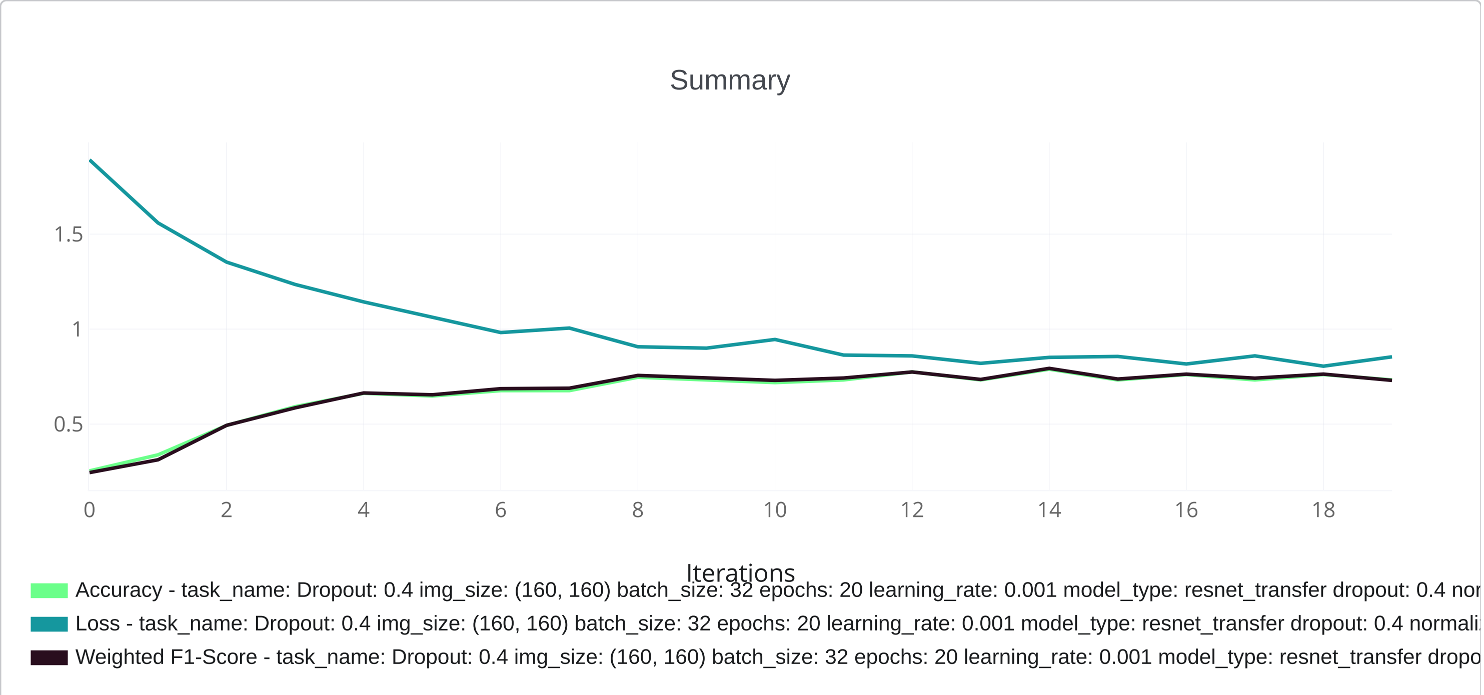
Dropout

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Dropout",
    "task_name":        "",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
```

```
    'add_newdata':      False

}


dropout = [0.0,0.2,0.4,0.8]
for dropouts in dropout:
  cfg['dropout'] = dropouts
  cfg['task_name'] = f"Dropout: {dropouts}"
  cfg["task_name"] = generate_taskname(cfg)
  start(cfg)
```



Dokładanie Danych

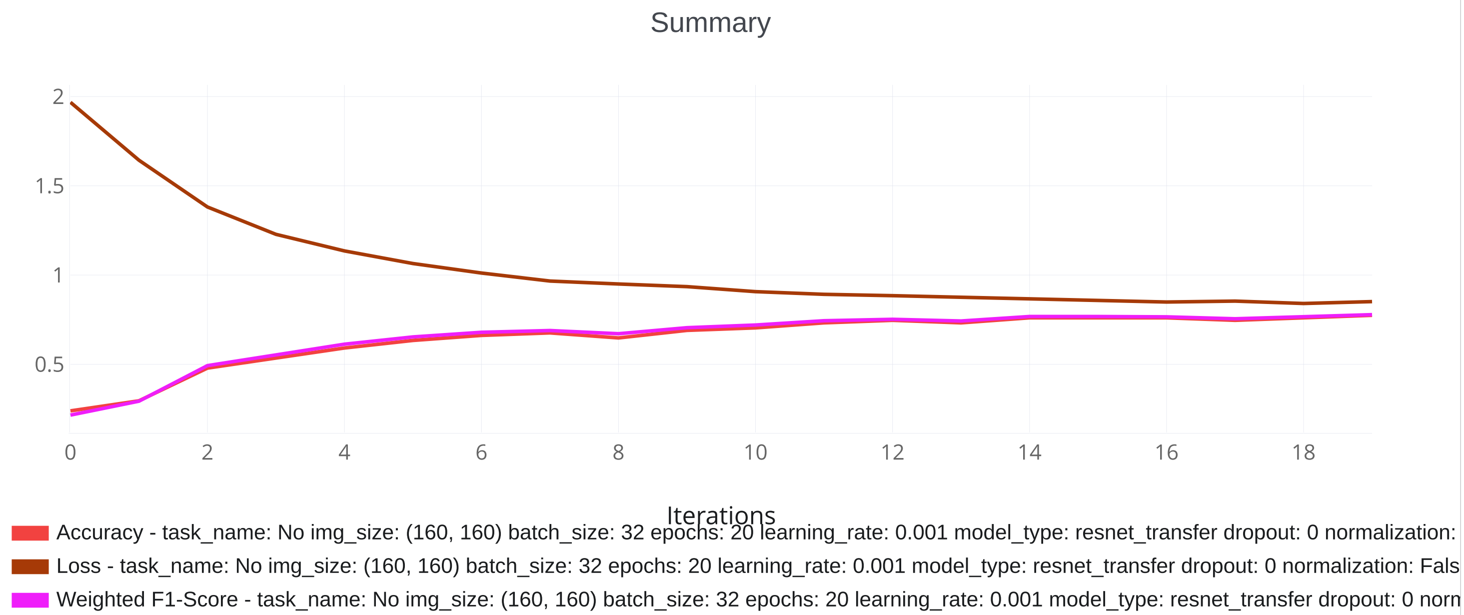
```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "AddData",
    "task_name":        "No",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
```

```
    'add_newdata':      False

}
cfg["task_name"] = generate_taskname(cfg)
start(cfg)

cfg = {
    "project_root":     "Projekt",
    "subproject":       "AddData",
    "task_name":        "Yes",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      True

}
cfg["task_name"] = generate_taskname(cfg)
start(cfg)
```



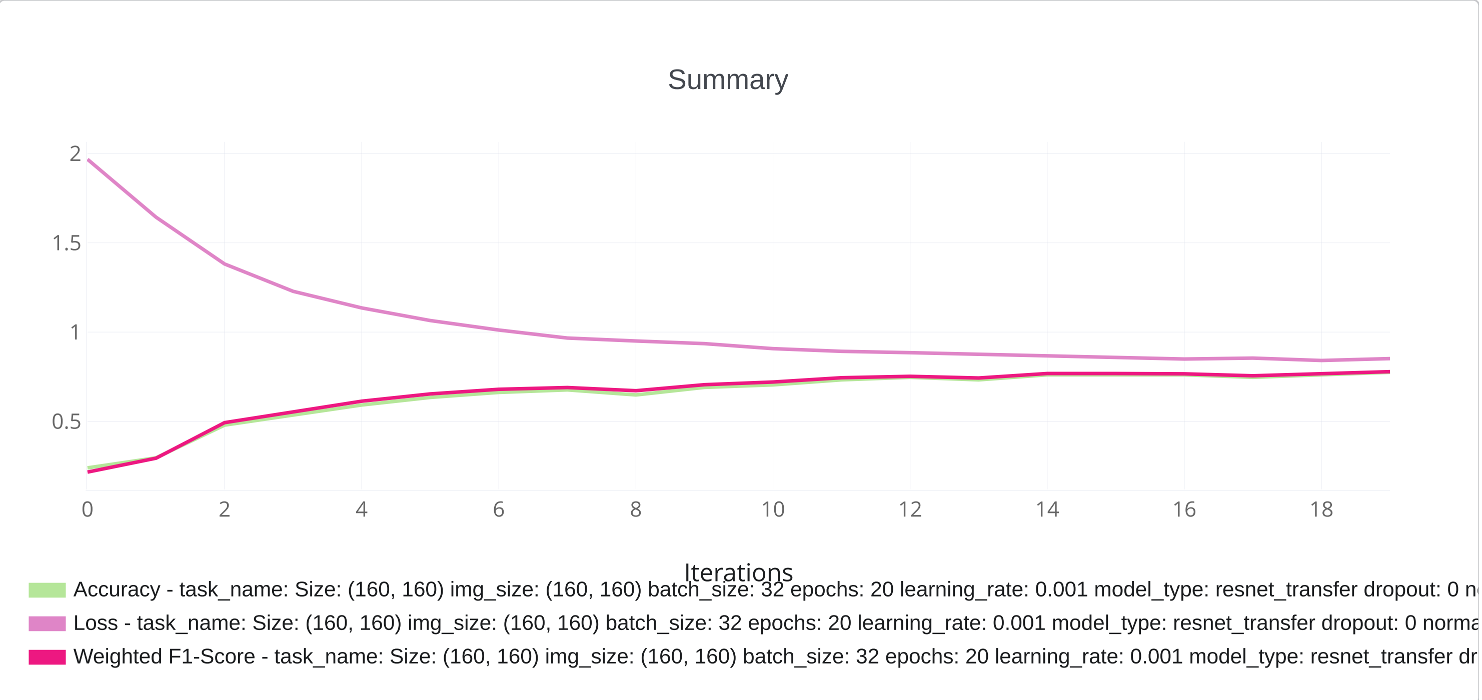
Testowanie Rozmiaru Obrazu

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "PhotoSize",
    "task_name":        "",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
```

```
    'add_newdata':      False

}


size = [(96, 96),(160, 160),(224,224),(360, 360)]
for sizes in size:
  cfg['img_size'] = sizes
  cfg['task_name'] = f"Size: {sizes}"
  cfg["task_name"] = generate_taskname(cfg)
  start(cfg)
```



Testowanie Rozmiaru Batcha

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Batch",
    "task_name":        "",
    "img_size":         (160, 160),
    "batch_size":       32,
    "epochs":           20,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0,
    'normalization':    False,
    'augment':          0,
```

```
    'add_newdata':      False

}


size = [8,16,32,64,128]
for sizes in size:
  cfg['batch_size'] = sizes
  cfg['task_name'] = f"Size: {sizes}"
  cfg["task_name"] = generate_taskname(cfg)
  start(cfg)
```

Szukanie najlepszego modelu

```
cfg = {
    "project_root":     "Projekt",
    "subproject":       "Final",
    "task_name":        "",
    "img_size":         (1000,1000),
    "batch_size":       8,
    "epochs":           25,
    "learning_rate":    0.001,
    "model_type":       "resnet_transfer",
    "dropout":          0.2,
    'normalization':    False,
    'augment':          0,
    'add_newdata':      True

}
```